

Joint Estimation of Thermal Conductivity and Interfacial Thermal Resistance in a Layered Slab Using Physics-Informed Neural Networks

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Draft status: This is a working manuscript. The introduction and methods describe the implemented benchmark and the intended comparative study. Inverse-PINN recovery, repeated training runs, and the complete noise comparison must be evaluated before publication claims are added.

1 Introduction

Heat transport through layered materials depends on both the thermal properties of the constituent layers and the resistance to heat flow at their interfaces. A finite interfacial thermal resistance allows a temperature discontinuity between adjacent materials even when heat flux is continuous. Consequently, neglecting imperfect thermal contact can alter the predicted temperature distribution and the interpretation of transient thermal measurements. A recent physics-informed neural network study explicitly addressed the heat equation under imperfect contact conditions, including the associated temperature jump and error analysis [3].

Estimating layer conductivity and interfacial resistance from temperature observations presents a coupled inverse problem. Both quantities influence the transient thermal response, and changes in one parameter may partially compensate for changes in the other. A close agreement between predicted and observed temperatures therefore does not, by itself, demonstrate accurate parameter identification. The reliability of an inverse method must also be assessed through parameter recovery, sensitivity to measurement noise, and consistency with the governing equations and interface conditions.

Physics-informed neural networks (PINNs) combine observations with residuals of physical equations in a common optimization problem. They can approximate a state field while simultaneously estimating unknown coefficients in its governing equations [1]. For layered heat conduction, this framework allows temperature observations to be combined with the heat equation, flux continuity, and the thermal-resistance jump condition. However, the representation of the temperature field requires care: a single globally smooth approximation cannot directly represent a finite temperature discontinuity at a material interface.

Domain decomposition provides a way to assign separate neural networks to different spatial or temporal regions. Space-time decomposition is an established extension of physics-informed learning [2]. In the present benchmark, the material interface and the termination of an imposed heating pulse provide natural locations for decomposition. The heating pulse also introduces abrupt changes in the exterior boundary flux, which a smooth network trained using boundary penalties may approximate inaccurately.

This study investigates joint identification of the thermal conductivity of a second material layer and the thermal resistance of its interface with a first layer. A one-dimensional synthetic benchmark is used so that the reference parameters, boundary conditions, and numerical data-generation process are explicitly controlled. The PINN uses separate temperature representations

for the two layers and for the heating and redistribution intervals. An analytical boundary lift imposes the prescribed exterior heat flux, while trainable corrections represent the finite-domain and interface response.

The intended comparison concerns parameter accuracy, temperature agreement, physics residuals, sensor placement, and sensitivity to noise. A conventional numerical least-squares inversion provides a reference method. The objective is to establish the conditions under which the PINN provides reliable identification, rather than to assume an advantage over conventional inversion. Because PINNs for imperfect-contact heat conduction already exist [3], the contribution must be established through the completed comparative experiments, not through the use of a PINN alone.

2 Materials and methods

2.1 Benchmark configuration and assumptions

The benchmark consists of two homogeneous layers arranged along the coordinate x :

$$\Omega_1 = (0, L_1), \quad \Omega_2 = (L_1, L_1 + L_2). \quad (1)$$

Heat transfer is assumed to be one-dimensional. Each layer has constant thermal conductivity and volumetric heat capacity. Internal heat generation, phase change, radiation, lateral heat loss, and temperature dependence of the material properties are excluded.

The interface has a constant areal thermal resistance R . Its heat capacity is neglected, so it stores no energy. The exterior right boundary is insulated, and the left boundary receives a prescribed rectangular heat-flux pulse.

The numerical properties in Table 1 define an illustrative benchmark. They do not represent a measured or specifically identified material pair.

Table 1: Properties and operating conditions of the synthetic layered-material benchmark.

Quantity	Symbol	Value
Layer 1 thickness	L_1	0.005 m
Layer 2 thickness	L_2	0.005 m
Known layer 1 conductivity	k_1	$10 \text{ W m}^{-1} \text{ K}^{-1}$
Reference layer 2 conductivity	k_2^*	$1 \text{ W m}^{-1} \text{ K}^{-1}$
Volumetric heat capacity, both layers	C_v	$2.0 \times 10^6 \text{ J m}^{-3} \text{ K}^{-1}$
Reference interface resistance	R^*	$10^{-3} \text{ m}^2 \text{ K W}^{-1}$
Initial temperature	T_0	293.15 K
Imposed pulse flux	q_0	10^4 W m^{-2}
Pulse duration	t_p	10 s
Reference simulation duration	t_{end}	200 s
Inverse observation interval		0.5–60 s

2.2 Governing equations

In layer i , the temperature $T_i(x, t)$ satisfies

$$C_v \frac{\partial T_i}{\partial t} = k_i \frac{\partial^2 T_i}{\partial x^2}, \quad i = 1, 2. \quad (2)$$

The heat flux is defined as positive in the direction of increasing x :

$$q_i = -k_i \frac{\partial T_i}{\partial x}. \quad (3)$$

The initial condition is

$$T_i(x, 0) = T_0. \quad (4)$$

At the heated exterior boundary,

$$q_1(0, t) = \begin{cases} q_0, & 0 < t < t_p, \\ 0, & t > t_p. \end{cases} \quad (5)$$

The right exterior boundary is insulated:

$$q_2(L_1 + L_2, t) = 0. \quad (6)$$

At the material interface, conservation of energy requires

$$q_1(L_1^-, t) = q_2(L_1^+, t) = q_\Gamma(t). \quad (7)$$

The interface temperature jump is

$$T_1(L_1^-, t) - T_2(L_1^+, t) = Rq_\Gamma(t). \quad (8)$$

Equation (8) is distinct from temporal continuity. The two material layers may have different temperatures at their interface, whereas temperature at a fixed location should remain continuous when the heating pulse terminates.

2.3 Synthetic reference data

Reference temperatures are generated using a cell-centered finite-volume discretization of Eq. (2). For adjacent cells j and $j + 1$, the thermal conductance per unit area is

$$G_{j+1/2} = \left[\frac{\Delta x_j}{2k_j} + R_{j+1/2} + \frac{\Delta x_{j+1}}{2k_{j+1}} \right]^{-1}, \quad (9)$$

where $R_{j+1/2} = R$ only at the material interface and is zero at all other internal faces.

The corresponding face flux is

$$q_{j+1/2} = G_{j+1/2}(T_j - T_{j+1}). \quad (10)$$

The cell energy balance is

$$C_v \Delta x_j \frac{dT_j}{dt} = q_{j-1/2} - q_{j+1/2}. \quad (11)$$

Backward Euler time integration gives

$$C_v \Delta x_j \frac{T_j^{n+1} - T_j^n}{\Delta t} = q_{j-1/2}^{n+1} - q_{j+1/2}^{n+1}. \quad (12)$$

The reference configuration uses 80 cells per layer and $\Delta t = 0.025$ s. Spatial and temporal refinement checks compare sensor temperatures across different cell counts and time steps. These comparisons quantify changes under refinement; they are not interpreted as exact discretization-error estimates.

Conservation is checked using the stored energy per unit area,

$$E_{\text{stored}}(t) = \sum_j C_v \Delta x_j [T_j(t) - T_0], \quad (13)$$

and the imposed energy,

$$E_{\text{input}}(t) = q_0 \min(t, t_p). \quad (14)$$

For the insulated benchmark, the long-time equilibrium temperature rise is

$$\Delta T_{\text{eq}} = \frac{q_0 t_p}{C_v (L_1 + L_2)} = 5 \text{ K}. \quad (15)$$

A conservative solver can still have discretization error. Energy conservation and temperature convergence are therefore evaluated separately.

2.4 Observation protocol and sensor layouts

Synthetic observations are extracted at

$$t_n = 0.5n \text{ s}, \quad n = 1, \dots, 120. \quad (16)$$

Two layouts are considered:

$$\mathcal{S}_{\text{same}} = \{2, 4\} \text{ mm}, \quad (17)$$

$$\mathcal{S}_{\text{across}} = \{2, 7\} \text{ mm}. \quad (18)$$

The first places both sensors in layer 1. The second places one sensor in each layer. Both layouts contain the same number of observations.

The current PINN pilot uses $\mathcal{S}_{\text{across}}$. The complete layout comparison requires running the PINN with both layouts under the same training protocol.

Noisy observations are defined by

$$y_{s,n} = T(x_s, t_n; k_2^*, R^*) + \varepsilon_{s,n}, \quad \varepsilon_{s,n} \sim \mathcal{N}(0, \sigma_T^2). \quad (19)$$

The conventional-inversion pilot uses $\sigma_T = 0.01 \text{ K}$ and 30 noise realizations. The shared 2 mm sensor receives the same noise realization in both layouts. The remaining sensor noises are independently drawn.

For a fair method comparison, the PINN and conventional inversion must use identical noisy observations. The completed conventional noise pilot does not constitute a completed PINN noise study.

2.5 Dimensionless formulation

For the equal-thickness layers, define local coordinates

$$z_1 = \frac{x}{L_1}, \quad z_2 = \frac{x - L_1}{L_2}, \quad \tau = \frac{t}{t_p}. \quad (20)$$

The dimensionless temperature rise is

$$u_i = \frac{T_i - T_0}{T_s}, \quad T_s = 5 \text{ K}. \quad (21)$$

Let

$$\kappa_i = \frac{k_i}{k_{\text{ref}}}, \quad k_{\text{ref}} = 1 \text{ W m}^{-1} \text{ K}^{-1}. \quad (22)$$

The dimensionless heat equation becomes

$$\frac{\partial u_i}{\partial \tau} - a_i \frac{\partial^2 u_i}{\partial z_i^2} = 0, \quad a_i = \frac{k_i t_p}{C_v L_i^2}. \quad (23)$$

For the benchmark,

$$a_1 = 2, \quad a_2 = 0.2\kappa_2. \quad (24)$$

The normalized rightward flux is

$$\hat{q}_i = \frac{q_i}{q_0} = -\frac{k_i T_s}{L_i q_0} \frac{\partial u_i}{\partial z_i}. \quad (25)$$

Thus,

$$\hat{q}_1 = -u_{1,z}, \quad \hat{q}_2 = -0.1\kappa_2 u_{2,z}. \quad (26)$$

The interface conditions become

$$\hat{q}_1(1, \tau) = \hat{q}_2(0, \tau), \quad (27)$$

$$u_1(1, \tau) - u_2(0, \tau) = \gamma \hat{q}_1(1, \tau), \quad \gamma = \frac{R q_0}{T_s}. \quad (28)$$

At the reference resistance, $\gamma = 2$.

2.6 Spatial and temporal network decomposition

Each material layer has two neural networks: one for $0 \leq \tau < 1$ and one for $1 \leq \tau \leq 6$. The implementation therefore contains four networks.

Each network has two inputs, three hidden layers containing 32 neurons each, and one scalar output. Hyperbolic tangent activation functions are used. Weights are initialized using Xavier normal initialization, with zero initial biases.

The time inputs are

$$\eta_H = 2\tau - 1, \quad \eta_C = 2\frac{\tau - 1}{5} - 1, \quad (29)$$

for heating and redistribution, respectively.

The spatial inputs are

$$\xi_1 = 2z_1^2 - 1, \quad \xi_2 = 2(1 - z_2)^2 - 1. \quad (30)$$

These transformations make the spatial derivative of the learned correction zero at the left exterior of layer 1 and the right exterior of layer 2. They do not prescribe the temperature at either exterior.

The heating corrections are

$$v_{i,H}(z, \tau) = g(\tau)\mathcal{N}_{i,H}(\xi_i, \eta_H), \quad g(\tau) = \frac{\tau}{\tau + 0.1}. \quad (31)$$

The redistribution corrections are

$$v_{i,C}(z, \tau) = \mathcal{N}_{i,C}(\xi_i, \eta_C). \quad (32)$$

The heating gate enforces a zero initial correction. The connection between the two time intervals is imposed through a temperature-continuity loss, rather than an exact constraint.

2.7 Analytical lift for the imposed heat-flux pulse

The left exterior flux is represented using a known solution of the dimensionless half-line heat equation. For $s > 0$, define

$$F(z, s) = 2\sqrt{\frac{a_1 s}{\pi}} \exp\left(-\frac{z^2}{4a_1 s}\right) - z \operatorname{erfc}\left(\frac{z}{2\sqrt{a_1 s}}\right), \quad (33)$$

and set $F(z, s) = 0$ for $s \leq 0$.

At regular points,

$$F_s - a_1 F_{zz} = 0, \quad (34)$$

and

$$F_z(0, s) = -1, \quad s > 0. \quad (35)$$

The pulse lift is

$$B(z, \tau) = F(z, \tau) - F(z, \tau - 1). \quad (36)$$

The predicted temperatures are

$$u_1(z, \tau) = B(z, \tau) + \begin{cases} v_{1,H}(z, \tau), & \tau < 1, \\ v_{1,C}(z, \tau), & \tau \geq 1, \end{cases} \quad (37)$$

and

$$u_2(z, \tau) = \begin{cases} v_{2,H}(z, \tau), & \tau < 1, \\ v_{2,C}(z, \tau), & \tau \geq 1. \end{cases} \quad (38)$$

Equations (30)–(38) enforce the imposed exterior flux and right insulation away from the pulse corner times. The initial temperature is also satisfied.

The half-line solution is used only as a boundary lift. It is not the solution of the finite two-layer problem. The learned corrections must still satisfy the finite-domain heat equations and material-interface conditions.

The lift depends on the known conductivity k_1 . It does not use the unknown k_2 or R .

Classical boundary derivatives are not assessed at the incompatible corner $(z_1, \tau) = (0, 0)$ or at the exact flux switch $(0, 1)$. Temperature predictions at the switch remain part of the observation protocol.

2.8 Trainable material properties

The unknown properties are represented by logarithmic parameters:

$$k_2 = k_{\text{ref}} \exp(\alpha), \quad R = R_{\text{ref}} \exp(\beta), \quad (39)$$

where $R_{\text{ref}} = 1 \text{ m}^2 \text{ K W}^{-1}$.

This parameterization guarantees positivity. It does not impose finite upper or lower bounds.

The initial inverse-PINN guesses are

$$k_{2,\text{init}} = 0.5 \text{ W m}^{-1} \text{ K}^{-1}, \quad R_{\text{init}} = 0.002 \text{ m}^2 \text{ K W}^{-1}. \quad (40)$$

Both properties are optimized jointly with the network weights and biases.

During fixed-parameter diagnostics, the properties are held at their reference values. Those diagnostics still use sensor data in the loss. They are therefore data-assisted field-fitting tests, not observation-free forward solutions.

2.9 Loss function

For sampled values $\{r_j\}_{j=1}^N$, define

$$\text{MSE}(r) = \frac{1}{N} \sum_{j=1}^N r_j^2. \quad (41)$$

The PDE residuals are

$$f_1 = u_{1,\tau} - 2u_{1,zz}, \quad f_2 = u_{2,\tau} - 0.2\kappa_2 u_{2,zz}. \quad (42)$$

The PDE loss is

$$\mathcal{L}_{\text{PDE}} = \text{MSE}(f_1) + \text{MSE}(f_2). \quad (43)$$

At the material interface, define

$$r_q = \hat{q}_1(1, \tau) - \hat{q}_2(0, \tau), \quad (44)$$

$$r_R = u_1(1, \tau) - u_2(0, \tau) - \gamma \hat{q}_1(1, \tau). \quad (45)$$

The interface losses are

$$\mathcal{L}_q = \text{MSE}(r_q), \quad \mathcal{L}_R = \text{MSE}(r_R). \quad (46)$$

For dimensionless observations $\tilde{y}_{s,n} = (y_{s,n} - T_0)/T_s$, the data loss is

$$\mathcal{L}_{\text{data}} = \sum_{s=1}^2 \text{MSE} [u_{i(s)}(z_s, \tau_n) - \tilde{y}_{s,n}], \quad (47)$$

where $i(s)$ identifies the layer containing sensor s .

The temporal connection loss is

$$\mathcal{L}_{\text{time}} = \sum_{i=1}^2 \text{MSE}[v_{i,C}(z, 1) - v_{i,H}(z, 1)]. \quad (48)$$

The analytical lift is shared by the two time branches and therefore cancels in this difference. The implemented total loss is

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + \mathcal{L}_q + \mathcal{L}_R + 10\mathcal{L}_{\text{data}} + 10\mathcal{L}_{\text{time}}. \quad (49)$$

Exterior boundary residuals are retained as diagnostics but excluded from Eq. (49), because the exterior flux conditions are imposed by construction.

Spatial and temporal derivatives are evaluated using automatic differentiation. All calculations use double precision.

2.10 Collocation sampling and optimization

Half of the interior collocation points are sampled uniformly in the heating interval $0 < \tau < 1$, and half in the redistribution interval $1 < \tau < 6$. Spatial coordinates are sampled uniformly in $0 < z < 1$. A small numerical exclusion avoids the exact pulse corner times.

This sampling gives equal numbers of points to intervals of unequal duration. The resulting loss is not a uniformly time-weighted integral over the complete observation interval.

The Adam stage uses 512 interior points per layer and 256 shared time samples for evaluating the material-interface conditions. Collocation points are refreshed every 100 steps. Temporal continuity is evaluated at 129 uniformly spaced spatial points in each layer.

Training starts with 5000 Adam steps at learning rate 10^{-3} . It is followed by L-BFGS with a maximum of 500 iterations, history size 50, and a strong-Wolfe line search.

During L-BFGS, the collocation set is fixed and contains 1024 interior points per layer and 512 interface time samples. The number of closure evaluations may exceed the number of optimizer iterations.

The pilot uses random seed 42 and an NVIDIA GeForce RTX 4050 Laptop GPU. Final reporting should include the installed software versions and results from repeated independent training seeds.

2.11 Conventional numerical inversion

A conventional least-squares inverse solver provides a comparison method. Its forward model uses the finite-volume spatial discretization with 60 cells per layer.

The semidiscrete system can be written as

$$\frac{d\boldsymbol{\vartheta}}{dt} = -\mathbf{A}\boldsymbol{\vartheta} + \mathbf{b}q_{\text{in}}(t), \quad (50)$$

where $\boldsymbol{\vartheta}$ contains cell temperature rises.

Modal decomposition of $\mathbf{A}$ gives the transient response for the prescribed rectangular pulse without backward-Euler time stepping. Sensor temperatures are obtained by interpolation from the cell values.

The estimated parameters minimize

$$\Phi(k_2, R) = \sum_{s,n} [T_{\text{num}}(x_s, t_n; k_2, R) - y_{s,n}]^2. \quad (51)$$

Optimization uses logarithmic parameters with bounds

$$0.1 \leq k_2 \leq 10 \quad [\text{W m}^{-1} \text{K}^{-1}], \quad (52)$$

and

$$10^{-5} \leq R \leq 10^{-2} \quad [\text{m}^2 \text{K W}^{-1}]. \quad (53)$$

The clean-data pilot uses three starting pairs:

$$(k_2, R) \in \{(0.5, 0.002), (2, 0.0002), (5, 0.005)\}, \quad (54)$$

with the units given above.

The conventional and PINN methods currently differ in their parameter bounds. This difference must be disclosed in comparisons of failure rates and robustness.

The reference generator and conventional inverse solver also use different spatial resolutions and time integration. Their discretization mismatch can introduce estimation bias, including for clean synthetic observations.

2.12 Accuracy, sensitivity, and physics diagnostics

Parameter recovery is measured using relative errors:

$$e_{k_2} = 100 \frac{\hat{k}_2 - k_2^*}{k_2^*}, \quad e_R = 100 \frac{\hat{R} - R^*}{R^*}. \quad (55)$$

Temperature agreement is quantified by

$$\text{RMSE}_T = \sqrt{\frac{1}{2N_t} \sum_{s=1}^2 \sum_{n=1}^{N_t} [\hat{T}(x_s, t_n) - y_{s,n}]^2}. \quad (56)$$

Because the current observations are used during training, this quantity measures agreement with fitted data. It is not a held-out prediction error.

Physics diagnostics use a fresh sample containing 2048 interior points per layer and 1024 interface time samples. PDE, interface-flux, interface-temperature-jump, and data residuals are reported separately.

The temporal connection is additionally evaluated on a 513-point spatial grid:

$$\Delta T_i^{\text{time}}(z) = T_s[v_{i,C}(z, 1) - v_{i,H}(z, 1)]. \quad (57)$$

Its RMS and maximum absolute values are reported for each layer.

Boundary-flux diagnostics are evaluated separately during startup, the heating plateau, the pulse switch, and subsequent redistribution. For the direct-flux representation, small exterior boundary errors are an architectural consequence and are not treated as independent evidence of successful field reconstruction.

Local parameter sensitivity is assessed using a Jacobian with respect to the logarithmic parameters:

$$J_{m\ell} = \frac{\partial T_m}{\partial p_\ell}, \quad \mathbf{p} = \left(\ln \frac{k_2}{k_{\text{ref}}}, \ln \frac{R}{R_{\text{ref}}} \right). \quad (58)$$

Singular values, the Jacobian condition number, and the cosine between its two sensitivity columns characterize local parameter coupling. These quantities provide local diagnostics and do not establish global identifiability.

For repeated noise realizations, parameter bias, standard deviation, and RMS estimation error are reported separately. A reduction in total estimation error is not interpreted as a reduction in noise sensitivity unless the spread also decreases.

2.13 Exploratory development and remaining validation

The development sequence examined a single time network, additional sampling near pulse transitions, temporal decomposition, an increased temporal-continuity weight, and the analytical exterior-flux lift.

These were exploratory changes made after inspecting pilot diagnostics. They are not presented as a preregistered comparison or as independent evidence of a generally optimal architecture.

Total loss values are not directly compared across configurations that change the sampling distribution, loss weights, or included terms. Common temperature errors and separately defined physics diagnostics provide the appropriate comparisons.

Before final conclusions are drawn, the study requires repeated inverse-PINN runs, matched noisy-data comparisons, both sensor layouts, and additional checks against reference temperatures away from the fitted sensors. Integrated energy balance and material-interface predictions should also be evaluated. A successful single clean-data inverse run is treated as a pilot result rather than a complete robustness demonstration.

3 Results

4 Discussion

5 Conclusions

Data and code availability

References

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